



LEARN DEEP: PLACEMENT PREPARATION

AI-Inspired Android Application for Structured Interview Mastery

· Android · Firebase · Kotlin

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A production-grade Android application combining AI-inspired recommendations, real-time analytics, structured quizzes, and personalized learning roadmaps — engineered for scale.

Interview Preparation Is Broken

Candidates face fragmented tools, zero personalization, and no measurable progress — leading to unstructured preparation and preventable failures.

1. No Structured Path



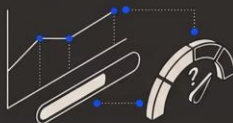
Candidates lack a guided learning roadmap.

2. Scattered Resources



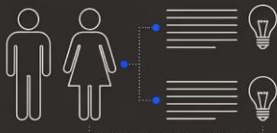
Content spread across dozens of disconnected platforms.

3. Zero Skill Tracking



No visibility into strengths, gaps, or progress over time.

4. No Personalization



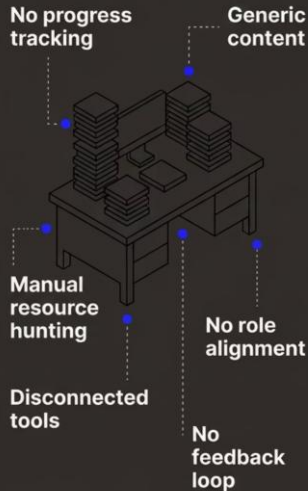
Generic advice with no role-specific guidance.

⚠ The Result: Most candidates walk into interviews underprepared — not from lack of effort, but from lack of direction.

Existing tools like LeetCode, Glassdoor, and YouTube are powerful in isolation but operate in silos. There is no unified platform that tracks skill growth, recommends personalized content, and measures readiness in real time.

Where Current Tools Fall Short

TRADITIONAL SYSTEMS



INTERVIEW PREP TRACKER



The Gap is Structural

Platforms like LeetCode focus on coding practice. LinkedIn Learning focuses on courses. Neither connects performance data to personalized next steps.

📌 Interview Prep Tracker closes this loop — turning raw practice into measurable, role-aligned readiness.

What Interview Prep Tracker Sets Out to Solve



Secure Authentication

Firebase-backed login and session management with role-aware onboarding.



Skill Analytics

Real-time dashboards tracking topic mastery, quiz performance, and growth trends.



AI Recommendations

Intelligent role-based suggestions powered by user performance data and Firestore.



Learning Roadmaps

Structured, curated paths aligned to target roles — frontend, backend, mobile, and more.



Adaptive Quizzes

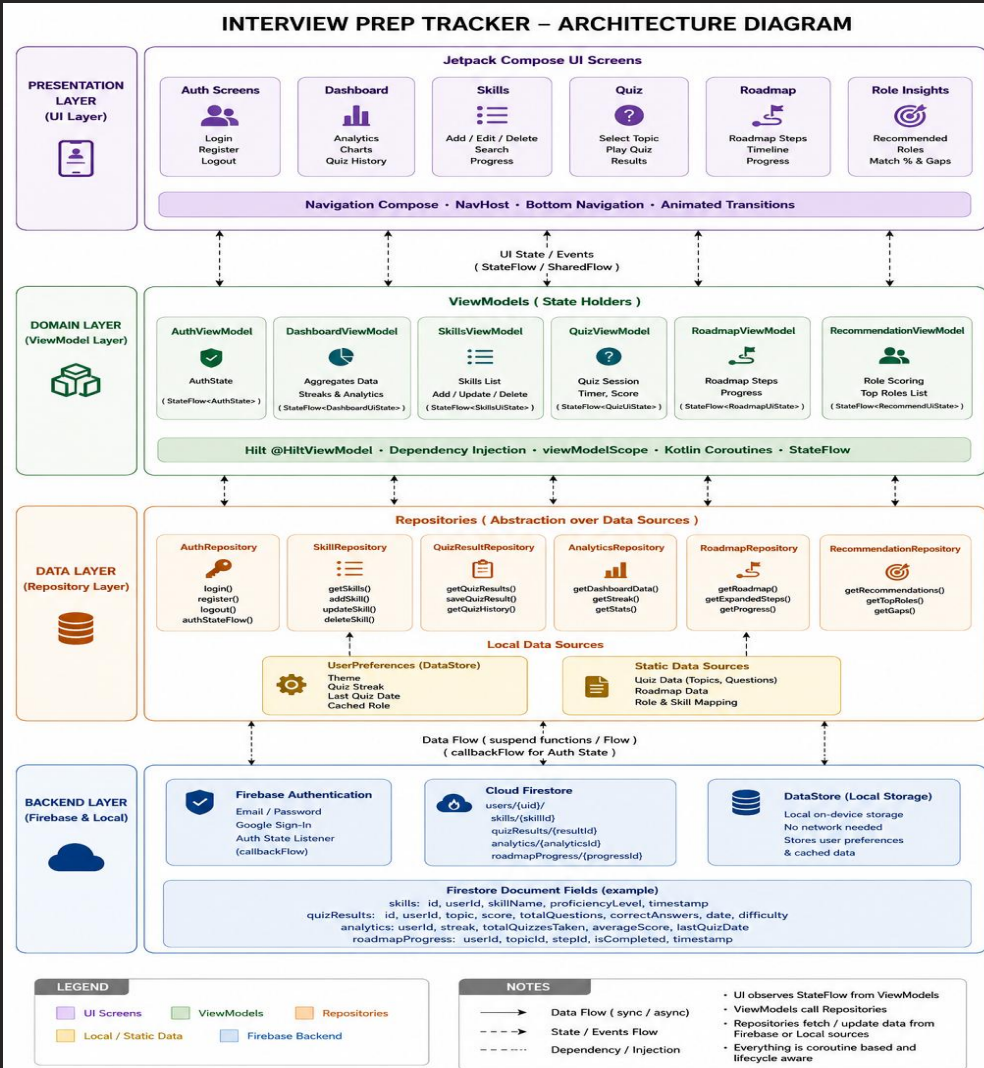
Topic-focused assessments with instant feedback and performance history.



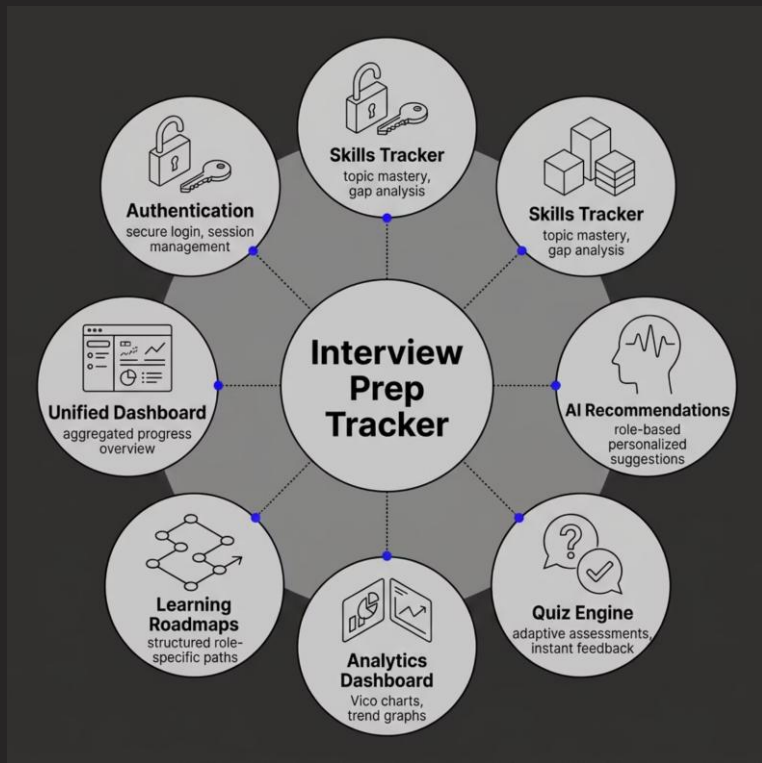
Unified Dashboard

A single command center aggregating all metrics, tasks, and progress at a glance.

Architecture Diagram



Core Feature Modules



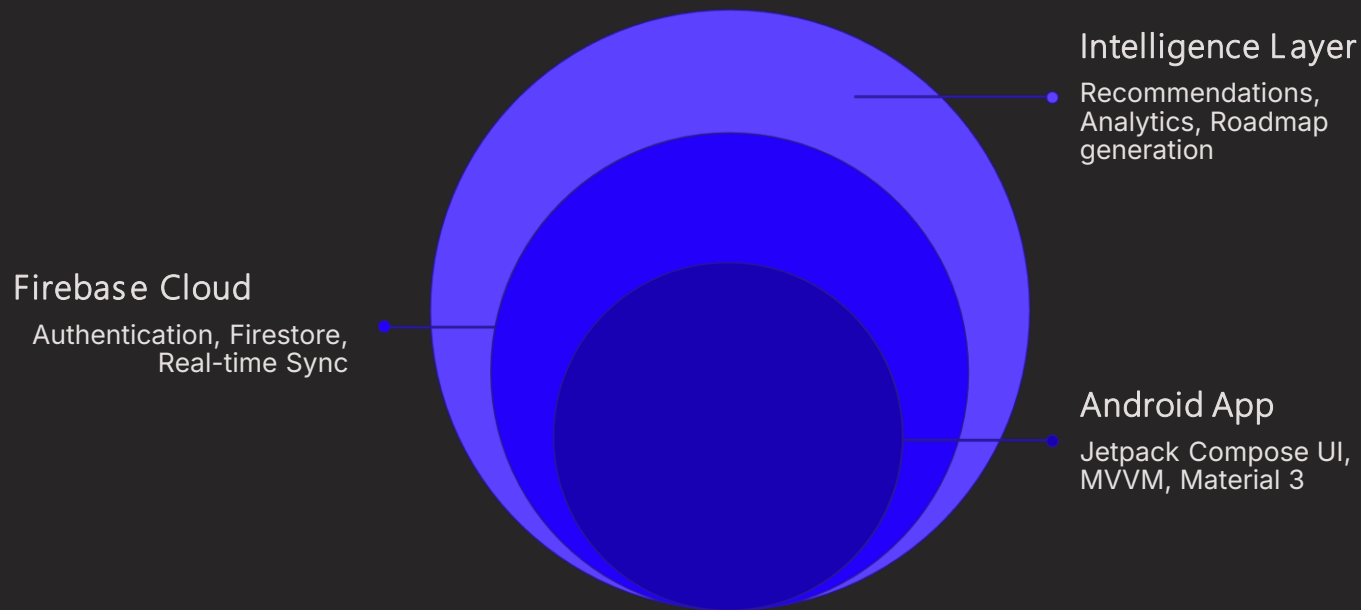
Modular by Design

Each feature module is independently scoped, Hilt-injected, and Compose-navigated — enabling clean separation and isolated testing.

- Navigation Compose manages the full backstack across all modules with type-safe argument passing and deep link support.

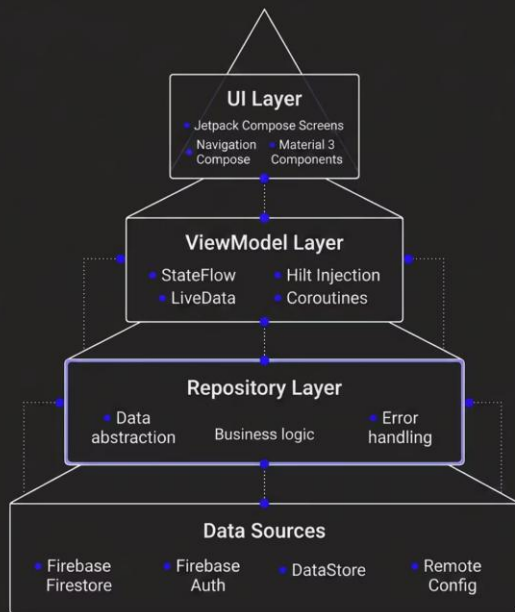
A Centralized Smart Ecosystem

Interview Prep Tracker unifies every preparation touchpoint into one intelligent, cloud-connected Android application.



The architecture enables bidirectional real-time data flow — user actions on-device instantly update cloud state, which feeds the recommendation engine and reflects back in the analytics dashboard.

Layered System Architecture



Separation of Concerns

Each layer has a single, well-defined responsibility. The UI never directly touches data sources — all state flows unidirectionally through ViewModels.

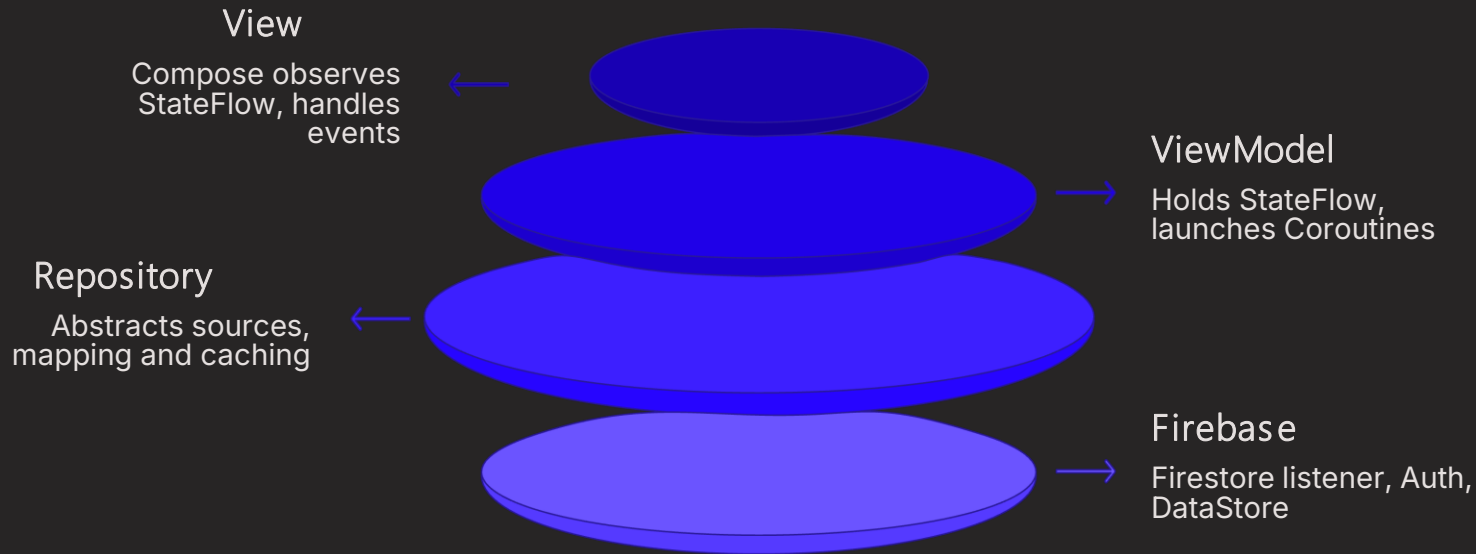
UI Layer

Compose screens observe StateFlow — pure reactive rendering.

Repository

Single source of truth, abstracting Firebase from business logic.

Clean Architecture with MVVM



StateFlow

Hot stream of UI state. Compose collects reactively — no manual lifecycle handling.

Coroutines

Structured concurrency for async Firebase operations without callback hell.

Hilt DI

Compile-time dependency injection. Scoped ViewModels, testable repositories.

Engineering-Grade Technology Stack



Kotlin

100% Kotlin codebase. Null-safety, extension functions, coroutine-native.



Jetpack Compose

Declarative UI with Material 3 theming and smooth animated transitions.



Firebase

Authentication, Firestore real-time DB, and Remote Config for cloud-driven features.



Hilt DI

Google-recommended DI built on Dagger. Scoped, testable, compile-safe.



Vico Charts

Beautiful, performant Compose-native charts for analytics dashboards.



DataStore

Jetpack DataStore replaces SharedPreferences for type-safe local preferences.

Skill Tracking Module

Full CRUD Operations

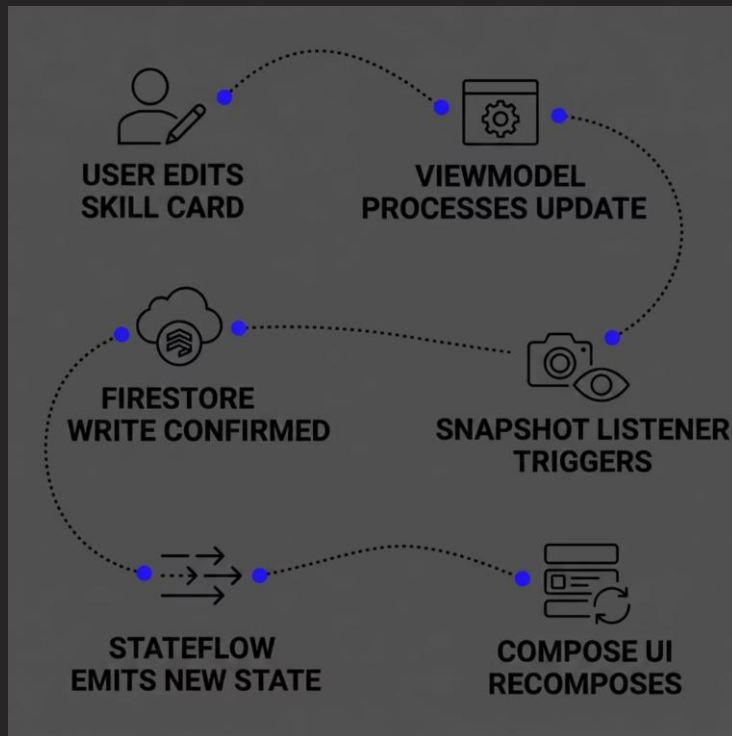
Create, read, update, and delete skills seamlessly. Each skill record carries a name, category, and proficiency level persisted in Firestore.

Proficiency Levels

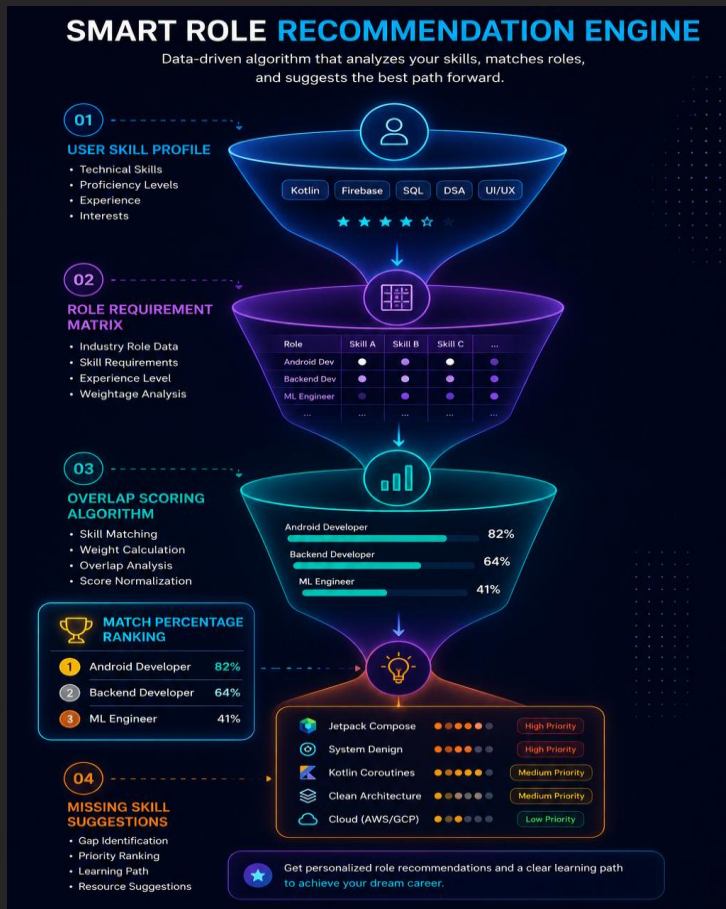
Skills are graded across four tiers — **Beginner** · **Intermediate** · **Advanced** · **Expert** — rendered as animated progress bars in Compose UI.

Real-Time Sync via Firestore

Snapshot listeners push delta updates instantly. UI state is managed through `StateFlow`, ensuring zero-latency reflection of cloud changes.



Role Recommendation Engine



An **AI-inspired matching system** that maps current skills against role requirements to surface personalized career paths.

Skill-Role Mapping

Each role carries a weighted requirement set. Matched against the user's proficiency matrix to produce a compatibility score.

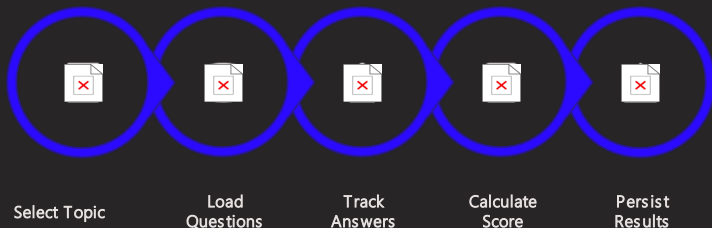
Match % Logic

Score = (matched skill weight sum) ÷ (total role weight) × 100. Displayed as a ranked card list with progress indicators.

Gap Analysis

Missing skills are flagged with priority labels, driving targeted roadmap suggestions directly tied to the recommendation result.

Quiz System



Questions are fetched per topic and difficulty tier. Scores are computed immediately post-submission and written back to Firestore for longitudinal tracking.



Topic-Wise Quizzes

Organized by domain — Data Structures, System Design, Kotlin, Android — with per-topic scoring history.



Difficulty Tiers

Easy · Medium · Hard. Adaptive selection surfaces harder questions as user performance improves over sessions.

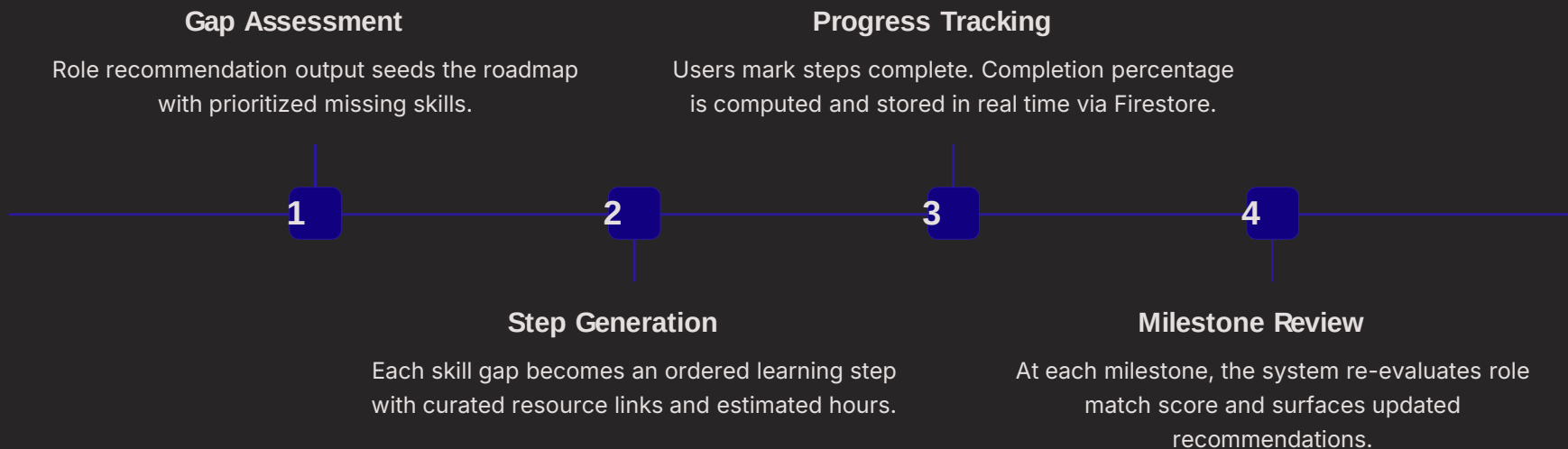


Result Persistence

Every attempt is stored with timestamp, score, and topic tag — enabling trend analysis inside the analytics dashboard.

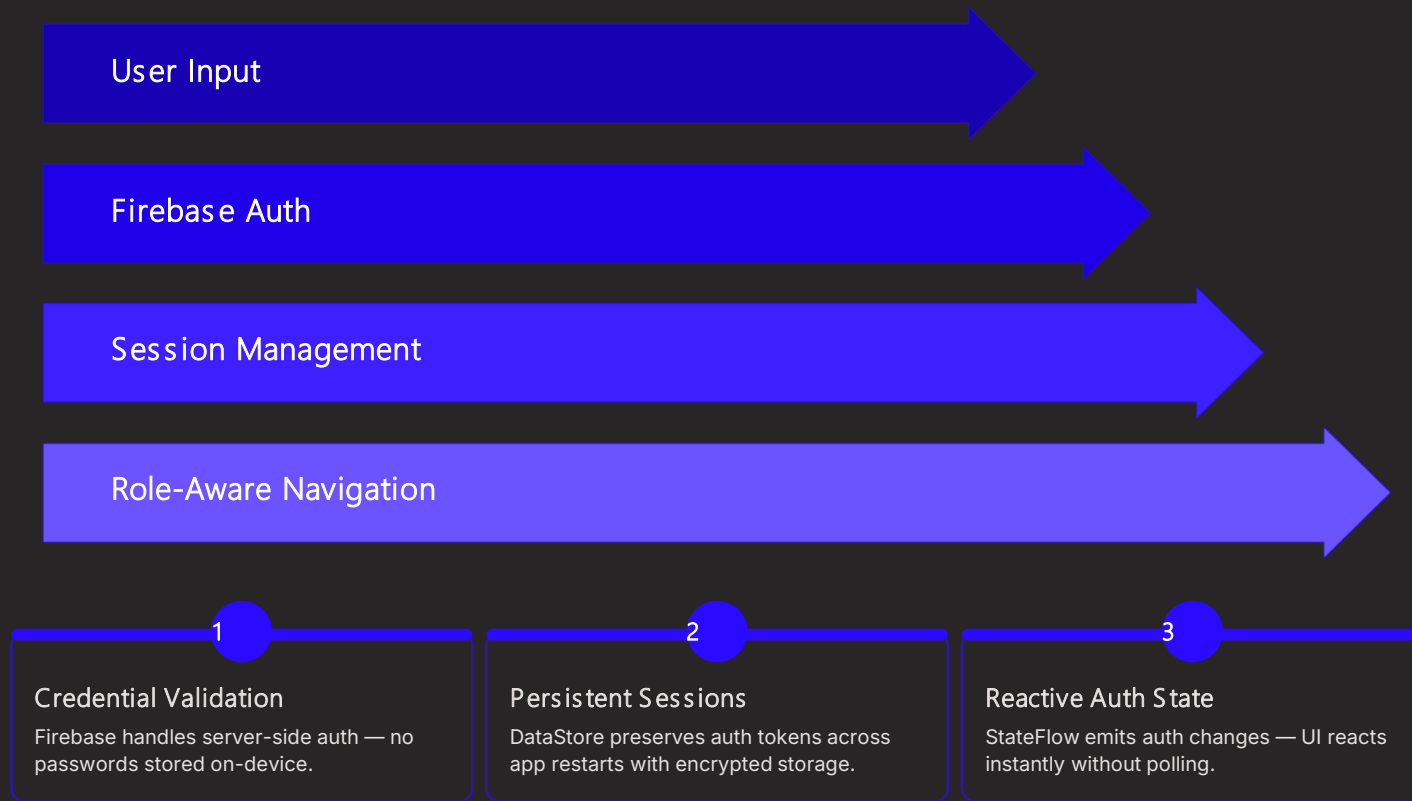
Roadmap System

Structured, milestone-driven learning paths generated from skill gaps and role targets — turning recommendation output into actionable guidance.

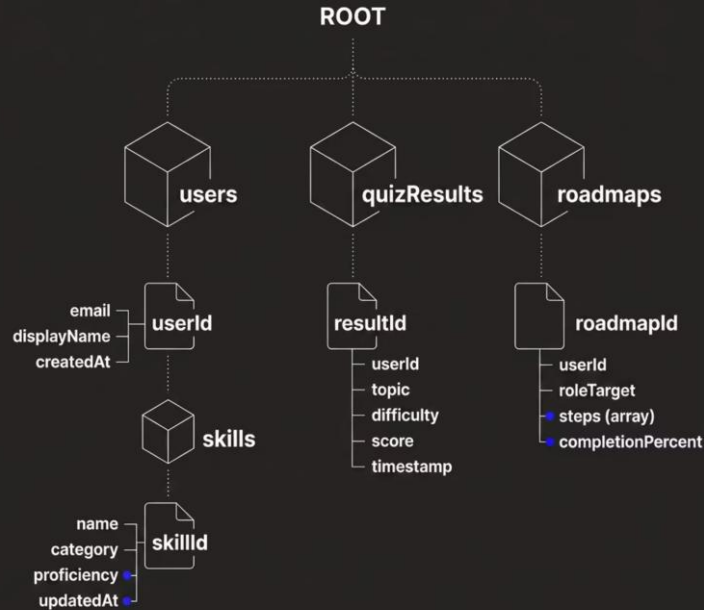


📘 Roadmap data is persisted in Firestore under the user's document, enabling cross-device continuity and historical milestone visualization.

Firebase Authentication Flow



Firestore Database Structure



users

Root user document. Holds profile metadata and serves as the parent for all user-scoped subcollections.

skills

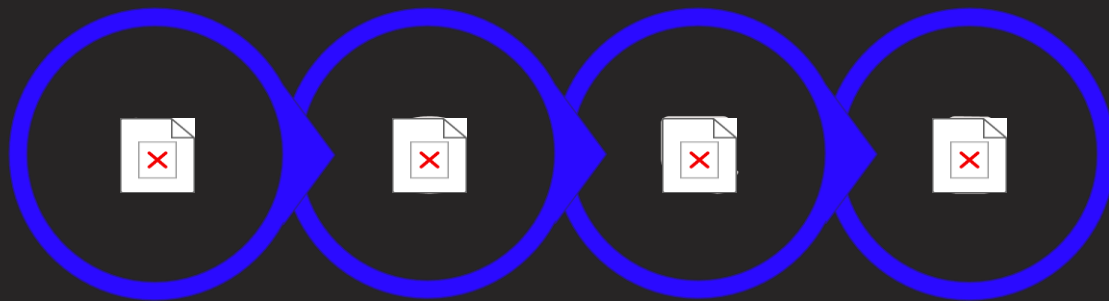
Subcollection under each user. Documents represent individual skills with proficiency and category fields.

quizResults

Top-level collection storing timestamped quiz attempts. Indexed by `userId` for efficient per-user queries.

📄 Security Rules enforce that each user can only read and write documents where `userId == request.auth.uid`.

Real-Time Data Flow



Snapshot
Listener

Repository
Map

ViewModel
Flow

Compose UI



Snapshot Listeners

Firestore `addSnapshotListener` streams document changes. No polling — pure push-based architecture.



Coroutines + Flow

Listeners are wrapped in `callbackFlow`, converting Firestore callbacks into cold Kotlin `Flow` streams.



StateFlow in ViewModel

Repository flows are collected into `MutableStateFlow`. UI observes via `collectAsStateWithLifecycle()`.



Scoped Lifecycle Safety

Coroutines launched in `viewModelScope` cancel automatically on ViewModel teardown, preventing memory leaks.

Security & Performance

Firebase Auth

Email/password and Google Sign-In via Firebase Authentication. JWT tokens managed transparently by the SDK.

Data Isolation

Firestore Security Rules enforce `request.auth.uid == userId`. No cross-user data access is architecturally possible.

Session Management

Auth state persisted via `DataStore`. Session invalidation triggers automatic `NavGraph` redirect to login screen.

PERFORMANCE OPTIMIZATION

- Async with Coroutines

All I/O operations run on `Dispatchers.IO`. UI thread is never blocked. Structured concurrency prevents leaks.

- Compose Smart Recomposition

`remember`, `derivedStateOf`, and `key()` minimize unnecessary recompositions. Stable data classes preserve Compose stability contracts.

- Hilt DI Efficiency

Singleton-scoped repositories and ViewModels injected via Hilt eliminate redundant object creation across the component lifecycle.

✓ The combined security and performance architecture ensures the app is production-ready — safe, fast, and maintainable at scale.

Future Enhancements

The roadmap extends Interview Prep Tracker into an intelligent, full-stack preparation platform.



AI-Powered Recommendations

On-device ML models (TensorFlow Lite) delivering personalized study paths based on performance history.



Interview Simulation Chatbot

Conversational AI conducts mock technical interviews, evaluates answers, and delivers structured feedback.



Resume Parser & Gap Analyzer

Upload a resume and receive role-fit scoring, missing skill identification, and targeted roadmap generation.



Gamification & ML Prediction

Streaks, badges, and leaderboards — paired with ML-based interview readiness score prediction per company target.

Thank You

Interview Prep Tracker

A modern Android engineering project — built with precision, designed for scale, and crafted for real-world impact.



Source Code

<https://github.com/230701257/InterviewPrepTracker1>



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Built with Kotlin · Kotlin · Firebase · Hilt · Material 3